Lung Cancer Treatment Navigator

Milestone 2: The Path to Precision

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The Problem: Evidence Overload

Oncologists cannot find the signal in the noise.

- **The Challenge:** Rapidly identifying new evidence for specific patient scenarios.
- **The Gap:** connecting evidence to current guideline contexts.
- **The Goal:** A defensible blueprint to retrieve, rank, and map medical evidence.

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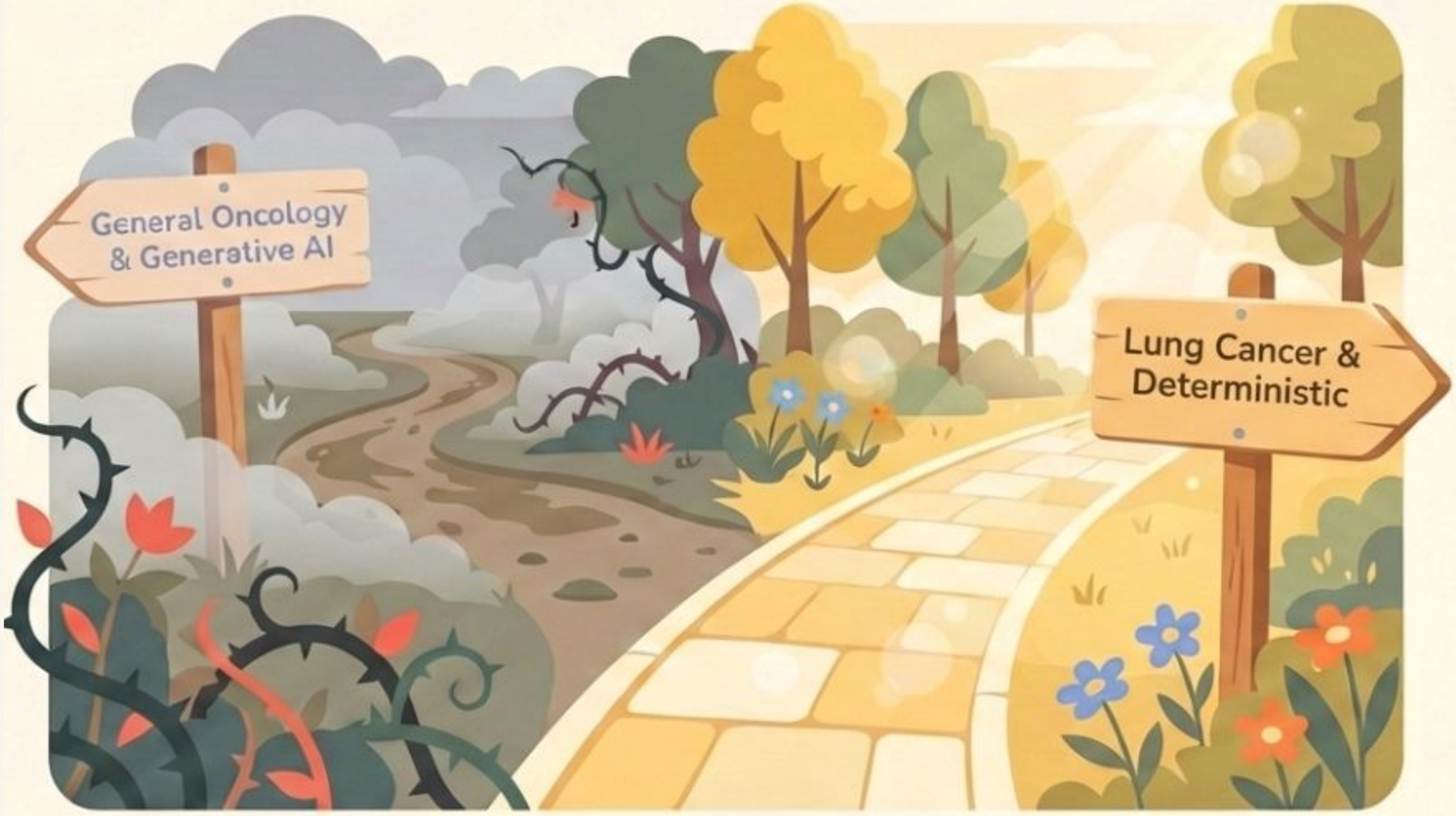
A Strategic Pivot to Precision



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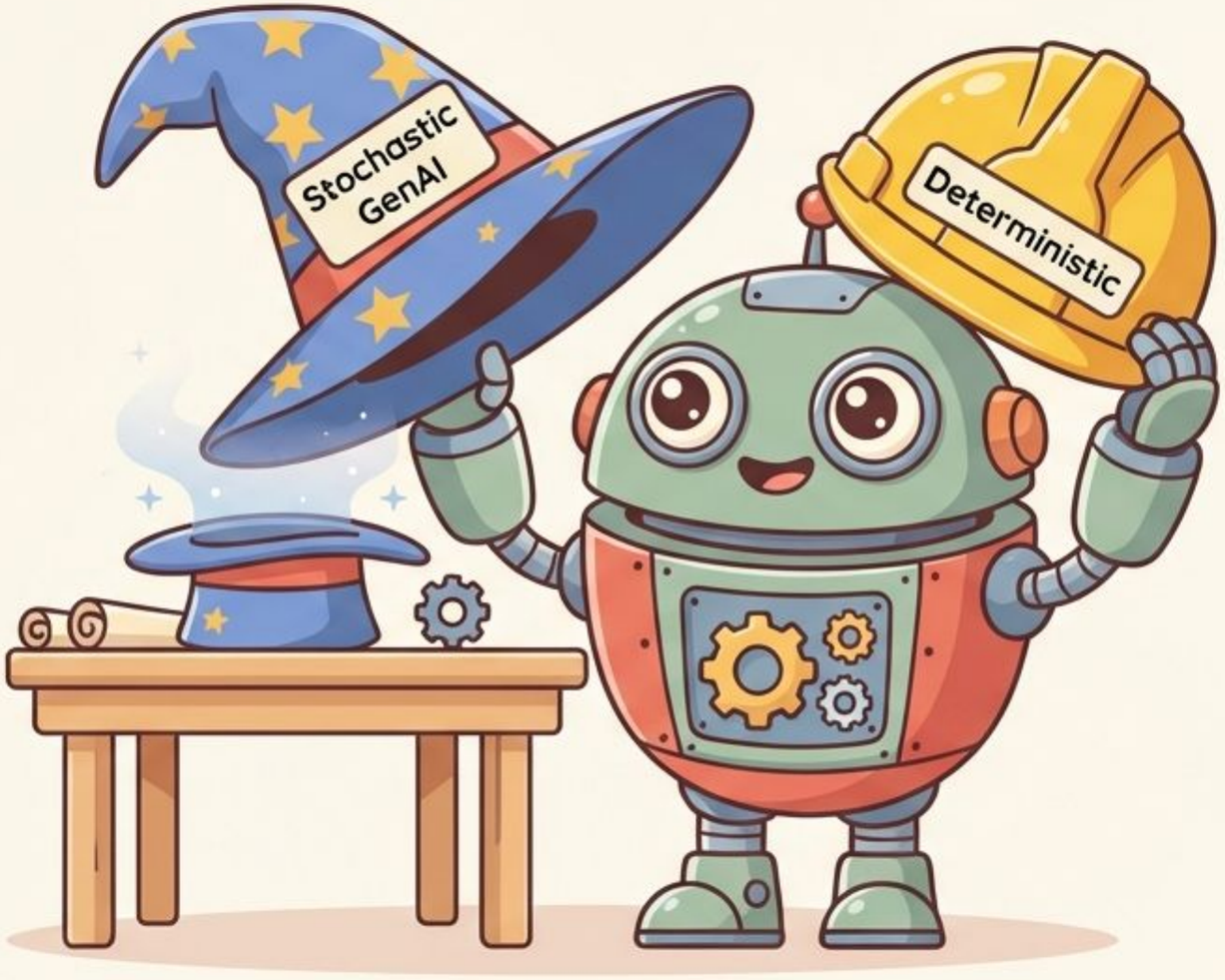
Replacing Wild Magic with Reliable Machinery

Moving from a creative AI Advisor to a Deterministic Rule-Based Engine.

- **Why?** To solve the challenge of evaluation.
- **The Result:** Verifiable, binary evaluation (Correct vs. Incorrect).
- **Role of LLMs:** Limited to bounded tasks like query expansion. Never medical reasoning.



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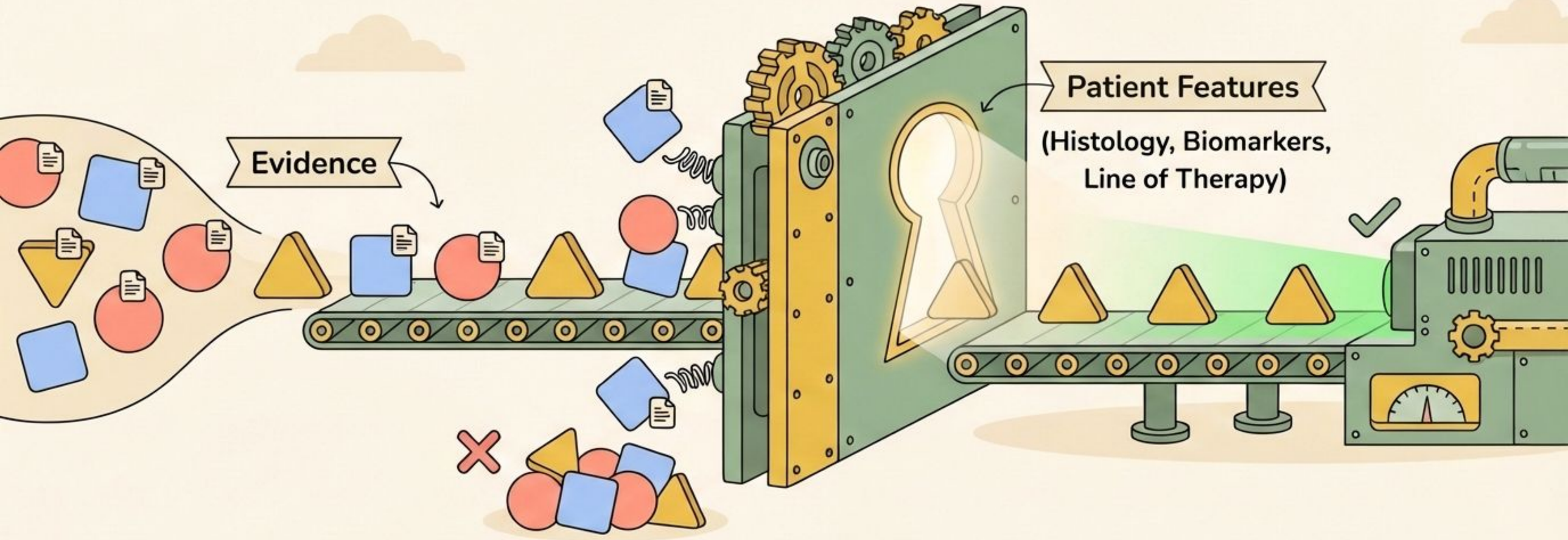
The Model: Relevance Gate



- Input: Structured Inputs only (No free text).
- Logic: If Evidence tags do not match the Patient Features, evidence is discarded.

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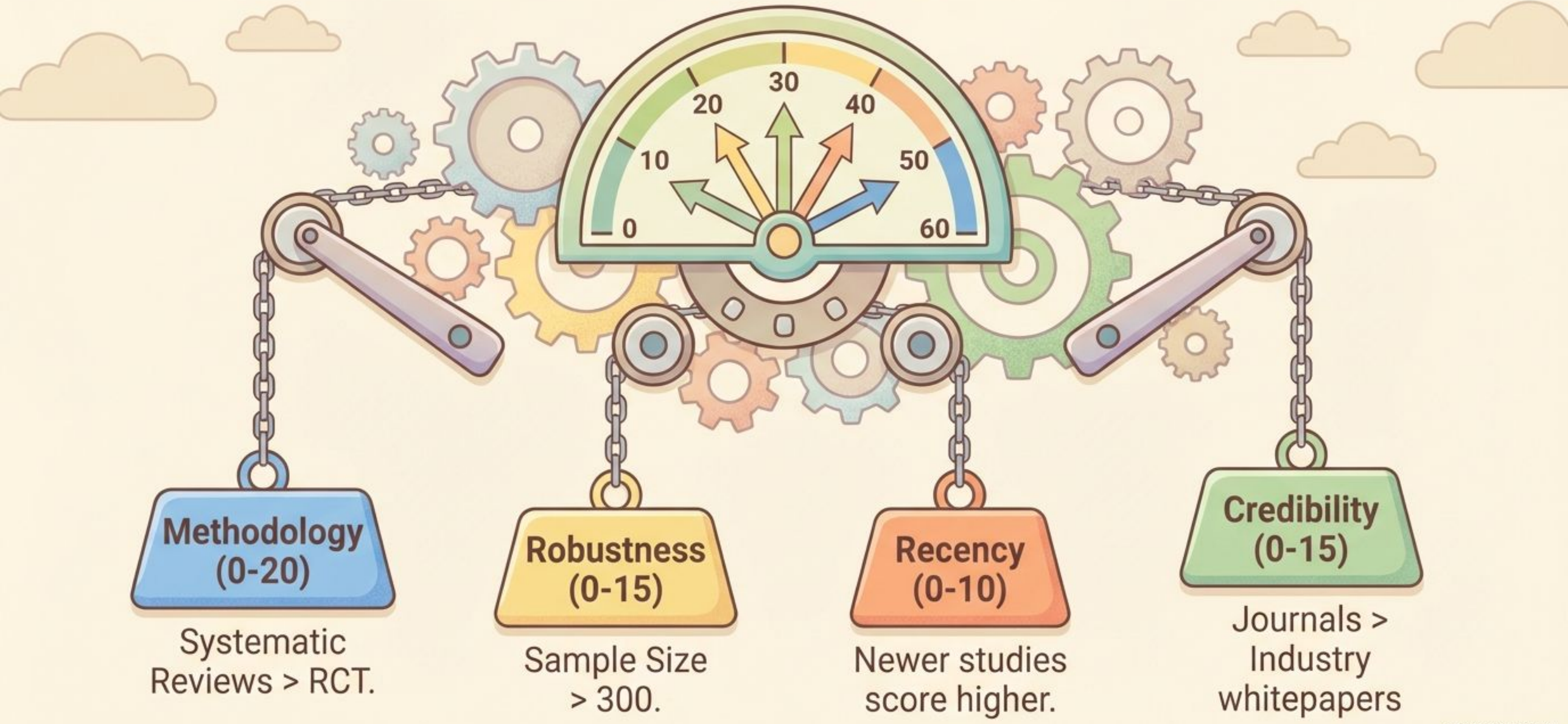
The Evidence Ranking Score (ERS)



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The Blueprint of the Machine

1. Retrieve

Relevant medical evidence.

2. Rank

Based on source quality and recency.

3. Map

Connect evidence to medical guidelines.

4. Label

Classify as Aligned, Silent, or In Conflict.

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Pillars of Truth

PubMed: The Library

Evidence Corpus.

ESMO: The Law

Guideline Truth.

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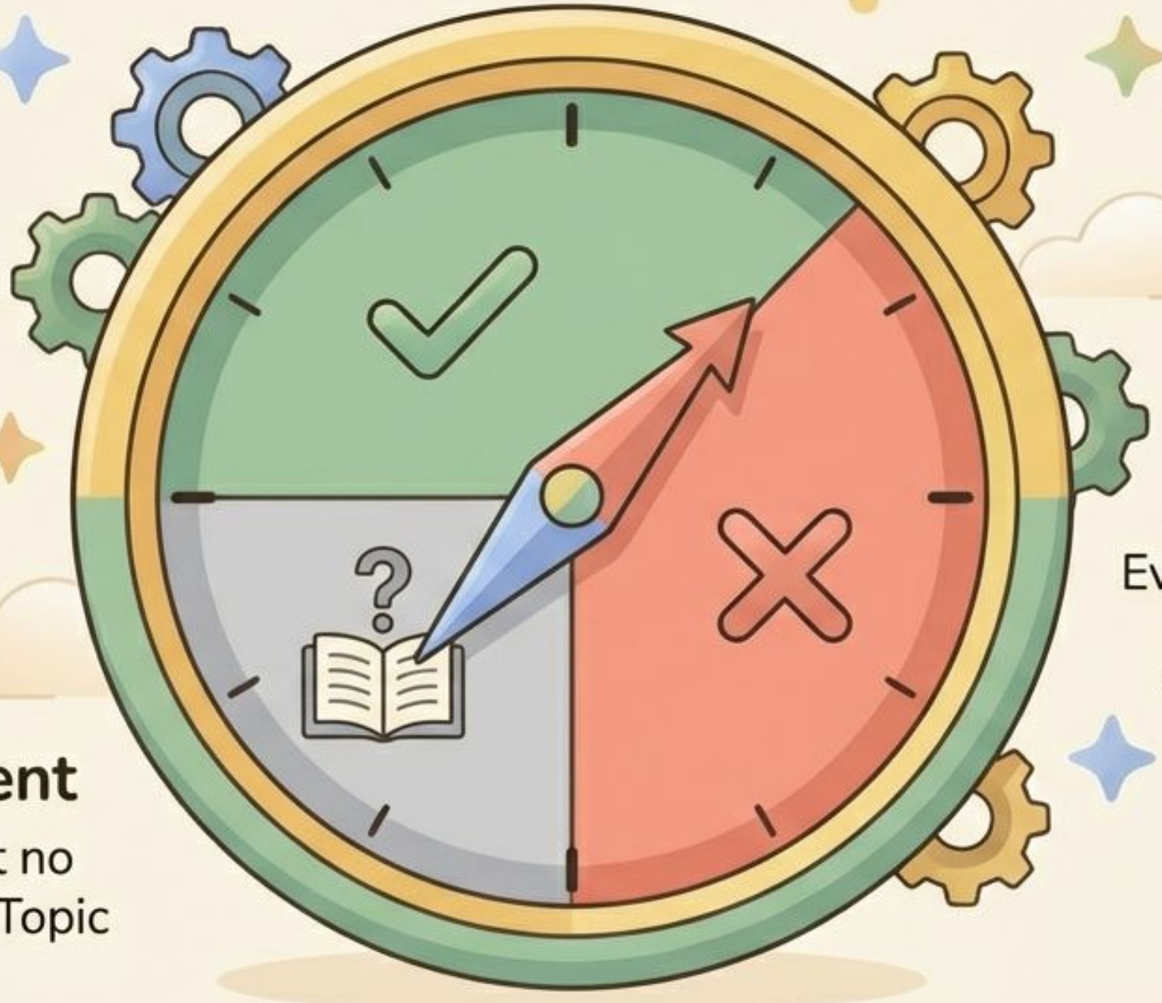
Navigating the Guidelines

Mapping Evidence to ESMO Stance



Aligned

Matches Guideline Topic & Stance

Conflict

Evidence suggests benefit where Guideline says "Do Not Recommend"

Guideline-Silent

Valid evidence, but no matching Guideline Topic

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The Gold Standard

15 Synthetic Patient Vignettes.

- **Purpose:** ground truth.
- **Structure:** fixed scenario + relevance + labels.

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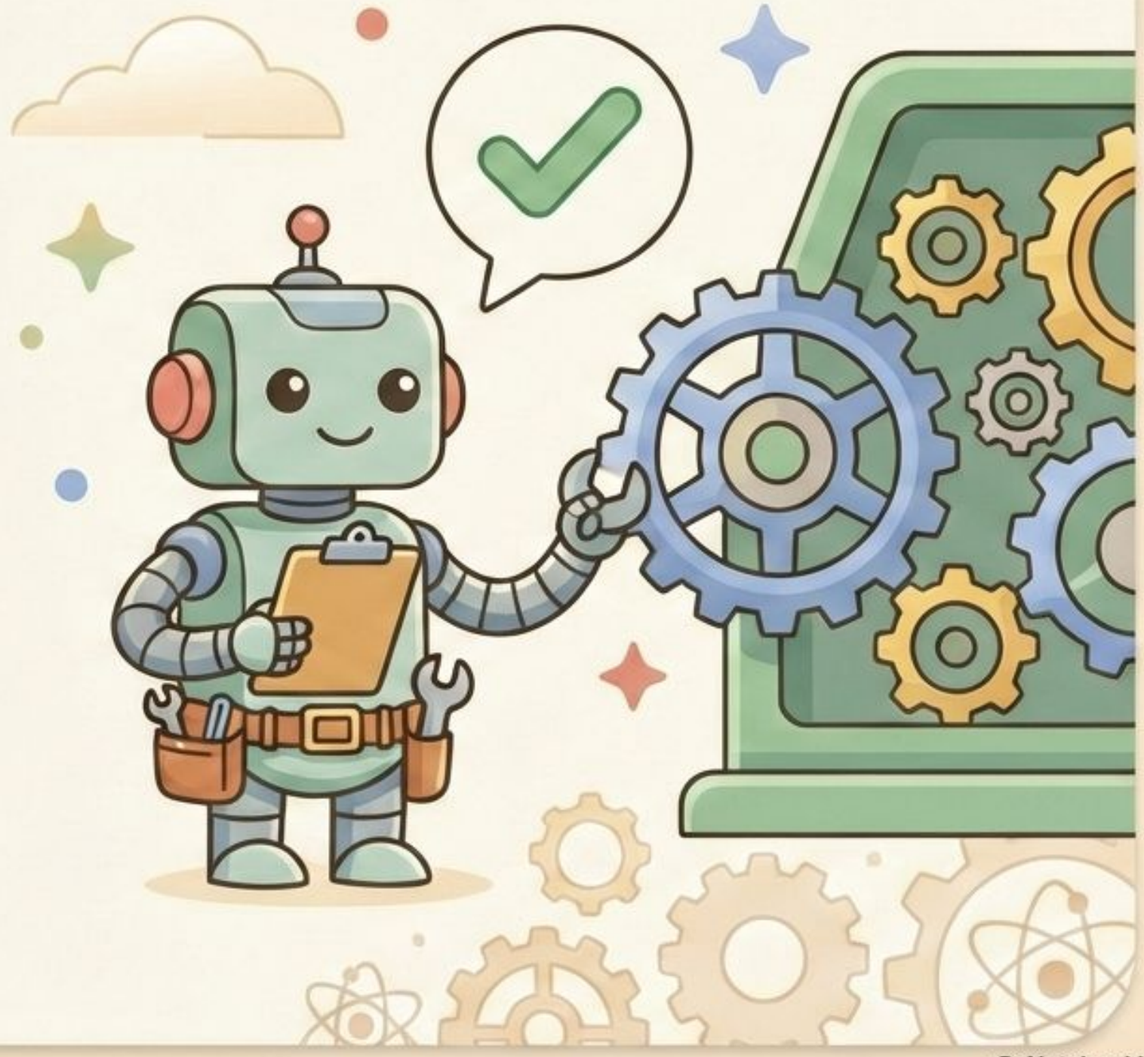
Evaluation Layer 1: System Integrity

- ✅ **Input Acceptance**
- ✅ **Successful Retrieval**
- ✅ **Valid Label Assignment**

Goal: Did the system run correctly?



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Evaluation Layer 2: Clinical Correctness

- **Evidence Relevance:** Target Recall ≥ 95% (Minimize False Negatives)
- **Guideline Mapping Error:** ≤ 10–15%
- **Citation Error Rate:** ≤ 10–15%

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The Non-Regression Promise

- **The Rule:** update requires re-scoring (Gold Vignettes).
- **The Consequence:** If scores drop, the update is blocked.

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The Road Ahead

Next Steps: Refining Data Extraction

- **PubMed Tagging:** Keyword Rules vs NLP.
- **Labeling:** Guideline Topic manual labeling.

Goal: Finalize reference mapping.

+ Live Demo! 💻

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Thank You!

By moving to **deterministic logic** and **limiting scope to NSCLC**, we ensure a tool that is not just smart, but **safe** and **auditable**.

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